

Exercise 02

Friday, 28 August 2026 11:48 pm

floating Decimals to binary conversion

① $(40.15625)_{10} = (\quad)_2$

2	40	R
2	20	0
2	10	0
2	5	0
2	2	1
2	1	0
2	0	1

bottom to top

$40 = (101000)_2$

$0.15625 \times 2 = 0.3125$	$= 0$	Top to bottom
$0.3125 \times 2 = 0.625$	$= 0$	
$0.625 \times 2 = 1.25$	$= 1$	
$0.25 \times 2 = 0.50$	$= 0$	
$0.50 \times 2 = 1.00$	$= 1$	

$(40.15625)_{10} = (101000.00101)_2$

► We only have a finite number of bits (digits) to store the mantissa

⇒ numbers that require more digits cannot be stored exactly

⇒ round-off error is of order smallest difference between numbers: 2^{p-m}

► How can we shorten "longer" mantissas to make them fit?

► 2 options:

Example (decimal): $\frac{20}{3} = 6.6666666\bar{6}$

a) **chopping**: simply discard the extra digits

if we only have 3 digits available: $\frac{20}{3} \approx 6.66$ ⇒ error $e = \frac{20}{3} - 6.66 = 6.\bar{6} \cdot 10^{-3}$

b) **rounding**: round the last available digit

if we only have 3 digits available: $\frac{20}{3} \approx 6.67$ ⇒ error $e = \frac{20}{3} - 6.67 = -3.\bar{3} \cdot 10^{-3}$

► In both cases we have made an error (round-off error): error = true value - numerical value

► this error doesn't seem like much, **BUT** if we perform a large number of operations,

round-off errors can accumulate and become large!

• Round-off Error

using Matlab:

Another example: $x = \frac{1}{a-b}$ with $a = 1.025432$ $b = 1.024567$ $x \approx 1.156069364161992 \cdot 10^3$

with chopping to 5 digits: $x \approx \frac{1}{1.0254 - 1.0245} \approx 1.1111 \cdot 10^3 \Rightarrow \text{error } e = 44.9582 \dots$

with rounding to 5 digits: $x \approx \frac{1}{1.0254 - 1.0246} \approx 1.2500 \cdot 10^3 \Rightarrow \text{error } e = 93.9306 \dots$

► What seems to be the problem here?

- Taking the difference of numbers that are similar $\Rightarrow$ round-off errors become important

• Truncation Error

► Taylor Series

$$f(x) = f(x_0) + (x - x_0) \left. \frac{df}{dx} \right|_{x=x_0} + \frac{1}{2!} (x - x_0)^2 \left. \frac{d^2f}{dx^2} \right|_{x=x_0} + \frac{1}{3!} (x - x_0)^3 \left. \frac{d^3f}{dx^3} \right|_{x=x_0} + \dots + \frac{1}{n!} (x - x_0)^n \left. \frac{d^n f}{dx^n} \right|_{x=x_0} + \dots$$

Example: $f(x) = \sin(x)$ with $x_0 = 0$:

$$\sin(x) = \sin(0) + (x - 0) \cos(0) + \frac{1}{2!} x^2 (-\sin(0)) + \frac{1}{3!} x^3 (-\cos(0)) + \frac{1}{4!} x^4 \sin(0) + \frac{1}{5!} x^5 \cos(0) + \dots$$

Example: $f(x) = \sin(x)$ with $x_0 = 0$: $\sin(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \frac{x^9}{9!} - \frac{x^{11}}{11!} + \dots$

Example: $\sin\left(\frac{\pi}{6}\right) = \frac{\pi}{6} - \frac{1}{3!} \left(\frac{\pi}{6}\right)^3 + \frac{1}{5!} \left(\frac{\pi}{6}\right)^5 - \frac{1}{7!} \left(\frac{\pi}{6}\right)^7 + \frac{1}{9!} \left(\frac{\pi}{6}\right)^9 - \frac{1}{11!} \left(\frac{\pi}{6}\right)^{11} + \dots$

only basic operations! But, need infinitely many terms!

In practice, we need to **truncate** the series! $\Rightarrow$ error is equal to all the terms we cut away

if we keep the first t terms, error is $e = \sum_{n=t+1}^{\infty} \frac{1}{n!} (x - x_0)^n \left. \frac{d^n f}{dx^n} \right|_{x=x_0}$

• Truncation Error

► Taylor Series

► for build-in functions like $\sin()$, $\cos()$, $\exp()$, etc., enough Taylor series terms are used so that the truncation error is comparable to the round-off error.

► In general, the truncation error depends on the specific numerical algorithm (here: how many terms we keep in the Taylor series)

► Note: the calculations of course also have round-off errors!

► for general numerical methods, truncation errors are usually much larger than round-off errors

- **Total Error** (aka **true error**)

- Definition: $\text{True Error} = \text{True Solution} - \text{Numerical Solution}$

- consists of truncation error & round-off error

- Definition: $\text{True Relative Error} = \frac{\text{True Solution} - \text{Numerical Solution}}{\text{True Solution}}$

- The above are straightforward, easy definitions, with one **MAJOR** problem:
we need to know the true solution!

- True (relative) errors are useful to test numerical methods

- For applications without true solutions, we need additional measures we will introduce throughout the class

$$f(x) = x^3 + 2x + 1, \quad g(x) = x^4.$$

Easy rule to remember

For polynomials,

$$f(x) = a_n x^n + \cdots + a_1 x + a_0,$$

the **highest power of x** determines the growth rate:

$$f(x) = O(x^n).$$

Consider

$$a_n = 3n^3 + 2n + 5, \quad b_n = n^4.$$

We want to determine the Big-O relationship.

Suppose

$$f(x) = x^2.$$

The exact derivative is

$$f'(x) = 2x.$$

We approximate the derivative using the **forward difference formula**:

$$f'(x) \approx \frac{f(x+h) - f(x)}{h}.$$

